Introduction to Internet of Things (IO4041)

Date: April 11th 2025

Course Instructor(s)

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Sessional-II Exam

Total Time (Hrs):	**1**
Total Marks:	**35**
Total Questions:	**3**

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Student Name	Roll No	Section	Student Signature

Instructions:

1. Attempt all questions on the answer booklet.
2. Write in blue or black ink only.
3. If you think some information is missing then make an assumption and state it clearly.

CLO 1: Discuss the basic IoT ecosystem, architecture and key components

Q1: MCQs – choose one correct answer, unless indicated otherwise. **[8 marks]**

1. Which technology requires purchase of a frequency band license
    a) LoRaWAN
    b) Zigbee
    c) Bluetooth
    d) Z-wave
2. What is the maximum number of active slave devices in a classic Bluetooth piconet?
    a) 3
    b) 5
    c) 7
    d) 15
3. Which Zigbee device type typically remains in sleep mode most of the time to conserve battery?
    a) Zigbee Coordinator (ZC)
    b) Zigbee Router (ZR)
    c) Zigbee End Device (ZED)
    d) Full Function Device (FFD)
4. Which features are common between Zigbee and Z-Wave protocols? (Select ALL that apply)
    a) Use CSMA/CA for channel access
    b) Operate in the 2.4 GHz band only
    c) Use IPv6 addresses for routing
    d) Employ mesh network topologies

5. Which of the following statements about LoRaWAN device classes are correct? (Select ALL that apply)
   a) Class A devices can transmit uplink messages at any time
   b) Class B devices use periodic receive windows
   c) Class C devices consume less power than Class A
   d) Class C devices have higher downlink latency than Class A
6. Which field of the IPv6 header is used to identify special handling for packets like real-time services?
   a) Payload Length
   b) Flow Label
   c) Hop Limit
   d) Next Header
7. In 6LoWPAN mesh-under routing, which type of addresses are used for routing?
   a) IPv4 addresses
   b) IPv6 addresses
   c) Link-layer addresses
   d) Home IDs
8. Write the abbreviated IPv6 address in its full form

   `2001:0:1::F:45C = 2001:0000:0001:0000:0000:0000:000F:045C`

CLO 3: Determine appropriate protocols and wireless connectivity technologies

Q2: Short answer questions **[3 + 5 + 2 + 4 marks]**

A. Differentiate between Bluetooth Classic and Bluetooth Low Energy (BLE) in terms of network topology.

Classic: one master and up to 7 inactive slaves form a piconet. All communication happens via master, who also controls medium access.

BLE: each master-slave relationship is a separate piconet.

B. Describe the routing method used by Z-Wave and the function of the 'healing' process.

Z-wave primarily uses **source routing** in which the senders include a packet's route within its header.

Initially all nodes send their neighbors information to primary controller, who is then able to build a network connectivity graph. Using this graph, the controlled computes all the routing table and disseminates to all router nodes. Later on, when a slave node sends a data packet to its parent (router), that will put the full route in packet header and forward it.

When there are connectivity changes in network, the precomputed routes might become invalid. Controller can initiate a heal process by sending heal command to all nodes, who recollect their neighbor information once again and forward to primary controller.

C. How is it that NFC tags are able to transmit their data without having a battery?

An NFC reader device emits a magnetic field which induces electricity in the nearby tag. That's how it gets powered and transmits data.

D. An organization is setting up an internal IoT network using IPv6. They need devices within the same sensor cluster (e.g., temperature sensors in one room) to communicate directly. They also need devices to communicate across different departments within the organization's network, but this traffic should never be sent to the public internet. Finally, some specific gateway devices need full internet reachability.
Describe how IPv6 address scopes would be used to meet these distinct requirements.

IPv6 defines three address scopes, each one meets specific requirements in given scenario.

Link Local address for communication within the subnet – direct communications within the same sensor cluster can use these addresses.

Unique Local address is meant for communication within the organization intranet – these will work best for sending data to other departments, since such packets are not forwarded by Internet routers.

Global addresses allow Internet reachability – these will be used by gateways for outbound traffic.

CLO 3: Determine appropriate protocols and wireless connectivity technologies

Q3: Short answer questions **[5 + 4 + 4 marks]**

A. An application generates a data payload of 470 bytes. This data is passed to UDP, which adds its standard 8-byte header. The resulting segment is then passed to IPv6 layer.
The IPv6 packet is then prepared with an unfragmentable hop-by-hop options header of 20 bytes and a fragmentable extension header of 25 bytes. These two are in addition to the standard IPv6 header.
The path from source to destination consists of four links, with MTUs of 320, 300, 750 and 560 bytes respectively.
Show the structure of all the fragments that will be created in order to send this data. For each fragment, clearly label the size of each header and payload block. Make sure to highlight the size of 'application data' in each fragment.

Path MTU = 300 bytes (minimum of all link MTUs)

Original packet structure is below.

IPv6 header 40	Hop-by-Hop 20	Ext header frag 25	UDP 8	App data 470

After fragmentation, an 8-byte fragmentation header FH is added in each fragment.

Fragment 1 (shaded payload is 232 — a multiple of 8)

IPv6 header	Hop-by-Hop	FH	Ext header frag	UDP	App data
40	20	8	25	8	199

Fragment 2 (shaded payload is again multiple of 8)

IPv6 header	Hop-by-Hop	FH	App data
40	20	8	232

Fragment 3 (last fragment, so shaded payload no longer needs to be a multiple of 8)

IPv6 header	Hop-by-Hop	FH	App data
40	20	8	29

B. For each of the following IoT applications, recommend one connectivity technology from the following: Z-wave, LoRaWAN, Bluetooth, RFID, 6LoWPAN. Give a brief but good justification for your choice, otherwise no credit will be awarded.

   i. A network of battery-powered sensors is deployed across a large farm, spanning several square kilometers. Each sensor monitors soil moisture or temperature, and sends a small data packet (few bytes) twice a day to a central farm management system. Sensors' battery should last multiple years.

   ii. A university library wants to implement a system to automate the book borrow/return process. Staff should also to be able to perform inventory checks by walking through aisles carrying a device. Gates at the exit should auto-detect any book stealing attempts.

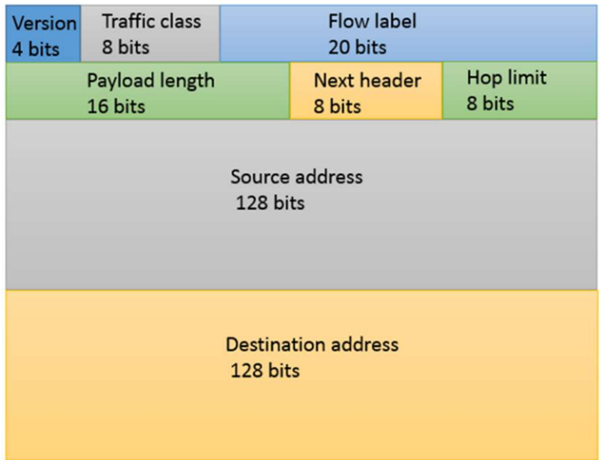
i.

LoRAWAN because sensor to gateway link can be several km, it supports large area of farm. Moreover only small amount of data is to be transmitted infrequently, which is as per LoRAWAN design goals.

ii.

RFID is suitable for this use case a tag can be pasted on each book. Reader devices can be installed at borrow/return station and exit door. Similarly, staff can carry a handheld reader while walking through aisles, and record which books are on shelves.

C. The 6LoWPAN adaptation layer uses header compression (IPHC) to reduce the significant overhead of the standard 40-byte IPv6 header. Explain which specific fields in the IPv6 header (shown below) offer the most significant opportunities for compression in a typical 6LoWPAN scenario (within the PAN communication) and how these fields can be compressed or omitted.



Version field can be omitted, it is always 6.

Traffic class and flow label together can be compressed to 2 bits by supporting only four traffic classes.

Payload length can be completely omitted, it can be deduced by receiver.

Hop limit can be compressed to 2 bits by supporting only three values.

Source & Dest IPv6 addresses can be fully omitted or compressed depending to scenario, such as direct neighbor communication (fully omitted) or known network prefix (first halves omitted).